

Complexity bounds for the rational Newton-Puiseux algorithm over finite fields

Adrien Poteaux · Marc Rybowicz

Received: 7 August 2009 / Revised: 24 February 2011 / Accepted: 18 April 2011 /
Published online: 6 May 2011
© Springer-Verlag 2011

Abstract We carefully study the number of arithmetic operations required to compute rational Puiseux expansions of a bivariate polynomial F over a finite field. Our approach is based on the rational Newton-Puiseux algorithm introduced by D. Duval. In particular, we prove that coefficients of F may be significantly truncated and that certain complexity upper bounds may be expressed in terms of the output size. These preliminary results lead to a more efficient version of the algorithm with a complexity upper bound that improves previously published results. We also deduce consequences for the complexity of the computation of the genus of an algebraic curve defined over a finite field or an algebraic number field. Our results are practical since they are based on well established subalgorithms, such as fast multiplication of univariate polynomials with coefficients in a finite field.

Keywords Puiseux series · Complexity · Algebraic functions · Genus

1 Introduction

1.1 Statement of the problem and motivations

Let K be a field and F be a polynomial of $K[X, Y]$ with degrees $\deg_X(F) = d_X > 0$, $\deg_Y(F) = d_Y > 0$ and total degree $\deg(F) = d$. We assume that F , considered as a

A. Poteaux (✉)
SALSA Project, INRIA, Paris-Rocquencourt Center, LIP6, UMR 7606,
Université Pierre et Marie Curie/CNRS, Paris, France
e-mail: adrien.poteaux@lip6.fr

A. Poteaux · M. Rybowicz
XLIM, UMR 6172, Université de Limoges/CNRS, Limoges, France
e-mail: marc.rybowicz@unilim.fr

polynomial in Y , is separable and primitive (i.e. it has no factor in $K[X]$). We denote by R_F the resultant of F and F_Y with respect to Y , where F_Y is the Y -derivative of F . Roots of $R_F \in K[X]$ are called *affine critical points* of F . The point at infinity is critical if $X = 0$ is a critical point of $F(1/X, Y)X^{d_X}$. Non critical points are called *regular points*.

Let x_0 be a critical point. If K has characteristic 0 or $p > d_Y$, each root of F , viewed as a univariate polynomial in Y , can be formally represented by a Laurent series S in $(X - x_0)^{1/e}$, called a *Puiseux series*, for a well-chosen positive integer e . The least such integer is the *ramification index* of S . Coefficients of S belong to a finite algebraic extension of K ; see Sect. 2.

Puiseux series are ubiquitous in the theory and practice of algebraic curves: For instance, they may be used to determine the genus of the curve $F(X, Y) = 0$ via Hurwitz formula [21], to compute integral bases of the function field $K(X)[Y]/(F)$ [45], to determine bases of $\mathcal{L}(D)$ spaces for divisors D by means of Dedekind-Weber's algorithm [2, 19], to approximate values of algebraic functions (when $K \subset \mathbb{C}$), etc.

A Puiseux series can be decomposed into two parts: the singular part, that captures important information such as ramification indices or Puiseux pairs, and the regular part; see Sect. 2. Let K' denote the field generated over K by the Puiseux series coefficients under consideration. Kung and Traub [26] showed that, once the singular part is known, the regular part can be computed using quadratic Newton iterations with $O(d_Y M(N))$ arithmetic operations in K' , where N is the number of terms required and $M(N)$ is the number of operations in K' necessary to multiply two polynomials of degree at most N in $K'[X]$. Since their purpose is to estimate the asymptotic complexity as a function of N , they do not study the complexity of the singular part computation, which is a constant independent of N . In practice, however, the singular part may already be a bottleneck.

In this paper, we focus on the complexity of the singular part, expressed as a function of the input size (namely, d_X , d_Y and the size of the coefficients of F), but also in terms of the output size.

From now on, assume that K is a field such that there exist algorithms for factorizing elements of $K[X]$. Singular parts of Puiseux series may be computed with the classical Newton-Puiseux algorithm [48] or the rational Newton-Puiseux algorithm introduced by Duval in [20]. The latter allows to take into account conjugacy over K and to restrict computations to *residue fields of places* (see Sect. 2), thus minimizing the required algebraic extensions of K and giving useful arithmetic information. It is therefore the method of choice for computer algebra systems; implementations are available in Maple [30], when K is an algebraic number field, (see the `algcurves` [puiseux] command due to Mark Van Hoeij [45]) and Magma ([3], command `RationalPuiseux`) for instance.

We became interested in the finite field case for the following reason: If K is an algebraic number field, both variants suffer from a dramatic coefficient swell that handicap their practical utility. Moreover they cannot be applied directly with floating point coefficients; see [32, 35] for examples and details. To overcome these problems, we have introduced a symbolic-numeric method:

- Important exact information, called *polygon trees*, is first obtained by means of Duval's algorithm applied modulo a well chosen prime number p . We gave a good reduction criterion for choosing a prime p such that F and $\overline{F} = F \bmod p$ have the same polygon tree. Proofs and details are given in [35]; see also [32]. We had to slightly modify classical Newton polygons to deal with cancellation modulo p of non essential coefficients. To this end, we introduced *generic Newton polygons*,¹ that we will also use in this paper, since they have other advantages; see Sect. 3. For instance, they provide *regularity indices* directly, and in terms of complexity, they correspond to worst-case situations.
- Then, polygon trees are used to guide floating point Puiseux series computations. A method was sketched in [31] and significant improvements, based on Singular Value Decompositions, were introduced in [32].

The rational Newton-Puiseux algorithm over finite fields is crucial in our strategy to obtain floating point Puiseux series; it is therefore important to understand its asymptotic behaviour.

Moreover, polygon trees encode ramification indices, Puiseux pairs, intersection multiplicity of branches, genus, and more. Complexity results for Puiseux series over finite fields, combined with our good reduction criterion and bounds for good primes p given in [35], lead to bit-complexity results for problems over algebraic number fields. We illustrate this approach in the conclusion with genus determination.

1.2 Contributions and contents

Let L be a finite field with $[L : \mathbb{F}_p] = t_0$, where p is a prime satisfying $p > d_Y$. In Sect. 2 we recall classical facts about rational Puiseux expansions and Puiseux series. Our variant of the rational Newton-Puiseux algorithm, based on generic Newton polygons and appropriate truncations of powers of X , is described in Sect. 3. Sections 4 and 5 contain our main contributions:

- First of all, we study truncation orders in terms of the output size (Propositions 1 and 2). We also show that truncation orders throughout the algorithm form a decreasing sequence. This justifies our variant of the algorithm; see Propositions 3 and 4.
- In Proposition 5, we prove that the output size is no greater than the X -adic valuation of the resultant R_F .
- Then, we study coefficient field representations. Since we need to change this representation along the course of the algorithm, we bound the number of operations in L required for this task. Again, we give results in terms of the output size (Proposition 6).
- For a fixed value of p , the asymptotic complexity is dominated by changes of variables; their cost is bounded in Proposition 7.

¹ Mark Van Hoeij pointed out to us that his implementation of the Newton-Puiseux algorithm, available since Maple V.5 (algcures[puisseux]), implicitly uses the concept of generic polygons. This code is used to compute integral bases [45], but this implementation technique has not been published.

- Unlike Duval [17], who relied on the D5 system to avoid factorization, we have chosen to factorize characteristic polynomials because efficient factorization algorithms exist over finite fields². Moreover, factorization provides a precise description of residue fields. Factorization costs are estimated in Proposition 8.
- Finally, we prove our **main result**, namely Theorem 3. Rational Puiseux expansions above the critical point 0 of a polynomial $F \in L[X, Y]$ may be computed with an expected number of :

$$O^{\sim} \left(d_Y^3 d_X^2 + d_Y^2 d_X t_0 \log p \right)$$

operation in L , where the notation $O^{\sim}$ hides logarithmic factors. This result was presented without proof at the ISSAC '08 conference [33] for a monic F and improves those of Duval, namely $O(d_Y^6 d_X^2)$ for a monic F and $O(d_Y^8 d_X^2)$ for the general case. It is worth noting that non monic cases require some care, since brute-force changes of variables to make polynomials monic do not yield the expected complexity.

- For expansions above *all* critical points of F , we have obtained a remarkably similar estimate, indicating that “most of the complexity” may be located above a single critical point; see Theorem 4. It also leads to Corollary 1: Provided that $p > d_Y$, the expected number of operations in L to **compute the genus of the algebraic curve** defined by $F(X, Y) = 0$ belongs to:

$$O^{\sim} \left(d_Y^3 d_X^2 t_0 \log p \right).$$

Surprisingly, we have not found complexity bounds for this problem in the literature.

- In the conclusion (Sect. 6), we give estimates for the complexity of the computation of the genus of a curve defined over an algebraic number field, using reduction modulo a prime p (Theorems 5 and 6). Again, we have not found similar results in the literature.

Efficient methods and implementations exist for the subalgorithms that we need over finite fields: multiplication, composition, and factorization of univariate polynomials, as well as primitive element computation and change of representation for finite fields. Hence, our algorithms can be effectively implemented within our complexity bounds and our results are of practical interest for accessible size problems. We emphasize that we have attempted to parametrize our bounds in terms of subalgorithm complexity and output size. Hence, it should be reasonably easy to update our results to take into account better algorithms for subproblems, or any a priori information about the output size.

² In particular, we have shown in [33, 35] that the size of L can be kept small when probabilistic versions of our symbolic-numeric method are used.

1.3 Other related works

Chistov [9] was first to show that the Newton-Puiseux algorithm over an algebraic number field had polynomial bit-complexity, but did not provide explicit exponents. Walsh [50] then showed that, for any $\epsilon > 0$, singular parts of Puiseux series for $F \in \mathbb{Q}[X, Y]$ can be computed using $O\left(d_Y^{32+\epsilon} d_X^{4+\epsilon} \text{ht}(F)^{2+\epsilon}\right)$ bit operations, where $\text{ht}(F)$ measures the size of the coefficients of F (see Sect. 6). He also proved that rational Puiseux expansions with polynomial size existed, but did not provide an explicit algorithm to compute them [49].

Other methods to compute Puiseux series have been proposed: [18], following [12], and [38]. We have explained in [35] why the Newton-Puiseux approach seems preferable in our context. No complexity results for these methods is known.

Merle and Henry [22] studied the arithmetic complexity of the resolution of the singularity at the origin defined by $F(X, Y) = 0$, assuming that F is irreducible in $K[[X, Y]]$. Using *lazy evaluation*, a programming technique that delays a calculation until the result is necessary to proceed further, they obtained an arithmetic complexity similar to Duval's, that is $O(d^8)$. Teitelbaum [42] considered a more general situation and gave a factorization-free algorithm that computes a representation for the resolution of a singularity defined by a bivariate power series. The arithmetic complexity, expressed in terms of a classical local invariant c_F called the *conductor degree* is $O(c_F^6)$. In general, c_F is not known in advance and would rather be a by-product of the resolution process. In our context, if F happens to be irreducible in $K[[X, Y]]$, c_F is smaller than the X -valuation of the discriminant of F with respect to Y [52], which in turn is bounded by $(2d_Y - 1)d_X$; this yields an upper a priori bound for the arithmetic complexity in $O(d^{12})$.

Finally, notice that Puiseux series may be computed efficiently in terms of the truncation order if a linear differential equation satisfied by the series is known [10, 11, 13, 43, 44]. But again, the singular part complexity has not been detailed, since these authors considered only the complexity as a function of the truncation order. Moreover, the computation of the differential equation may be a bottleneck since it usually has coefficients with fairly large degrees: a bound in $O(d_X d_Y^4)$ was given in [15] and reduced to $O(d_X d_Y^3)$ in [4], but only when K has characteristic zero.

1.4 Notations

- If L is a field, $\overline{L}$ will denote an algebraic closure of L .
- If $H \in L[X, Y]$, then H_X and H_Y are the formal partial derivatives of H . We denote by $\deg_X(H)$ (resp. $\deg_Y(H)$, $\deg(H)$) the degree of H with respect to X (resp. Y , resp. the total degree of H). The leading coefficient of H with respect to the variable, say Y , is represented by $\text{lc}_Y(H)$.
- We shall say that H as above is *separable* if it is separable viewed as a univariate polynomial in Y .
- For each positive integer e , ζ_e is a primitive e th root of unity in $\overline{L}$. Primitive roots are chosen so that $\zeta_{ab}^b = \zeta_a$.

- v_X denotes the X -adic valuation of the fractional power series field $L((X^{1/e}))$, normalized with $v_X(X) = 1$. If $S \in L((X^{1/e}))$, we denote by $\text{tc}(S)$ the trailing coefficient of S , namely $S = \text{tc}(S)X^{v_X(S)} + \text{higher order terms}$.
- If e is a positive integer, the Galois group of $\bar{L}((X^{1/e}))$ over $\bar{L}((X))$ is denoted by $\mathbb{G}_e$. It is the cyclic group generated by the automorphism $X^{1/e} \mapsto \zeta_e X^{1/e}$.
- If $S = \sum_k \alpha_k X^{k/e}$ is a fractional power series in $L((X^{1/e}))$ and r is a rational number, $\lceil S \rceil^r$ denotes the truncated series $\lceil S \rceil^r = \sum_{k \leq N} \alpha_k X^{k/e}$ where $N = \max\{k \in \mathbb{N} \mid \frac{k}{e} \leq r\}$. We generalize this notation to elements of $L((X^{1/e}))[Y]$ by applying it coefficient-wise. In particular, if $H \in L[[X]][Y]$ is defined as $H = \sum_i (\sum_{k \geq 0} \alpha_{ik} X^k) Y^i$, then $\lceil H \rceil^r = \sum_i \left(\sum_{k=0}^{\lfloor r \rfloor} \alpha_{ik} X^k \right) Y^i$.
- If U is a univariate polynomial, then Δ_U denotes the discriminant of U and R_U denotes the resultant of U and its derivative. If U is a multivariate polynomial, the context will always allow to identify the relevant variable.

2 Rational Puiseux expansions

The purpose of this section is to recall facts about classical Puiseux series and rational Puiseux expansions. For short, they will respectively be called CPS and RPE.

Let L be a field of characteristic $p \geq 0$ and H be a separable polynomial in $L[X, Y]$, primitive in Y , with $\deg_Y(H) > 0$. We assume that L and H satisfy the characteristic condition:

$$p = 0 \quad \text{or} \quad p > \deg_Y(H) \quad (1)$$

Up to a change of variable $X \leftarrow X + x_0$ or $X \leftarrow 1/X$, we suppose that $X = 0$ is a critical point and we will consider CPS and RPE above 0.

Definition 1 A parametrization $R(T)$ of H is a non constant pair of power series $R(T) = (X(T), Y(T)) \in \bar{L}((T))^2$ such that $H(X(T), Y(T)) = 0$ in $\bar{L}((T))$. The parametrization is *irreducible* if there is no integer $u > 1$ such that $R(T) \in \bar{L}((T^u))^2$. The *coefficient field* of $R(T)$ is the extension of L generated by the coefficients of $X(T)$ and $Y(T)$.

For the convenience of the reader, we recall classical definitions [8, 27]:

Definition 2 Assume H irreducible in $L[X, Y]$. Let $\mathcal{K}_H = L(X)[Y]/(H)$ be the algebraic function field defined by H . A *valuation ring* $\mathfrak{O}$ of $\mathcal{K}_H$ is a proper subring of $\mathcal{K}_H$ such that, for any non zero element $z \in \mathcal{K}_H$, either $z \in \mathfrak{O}$, or $1/z \in \mathfrak{O}$; it is a local ring. A *place* of $\mathcal{K}_H$ is the unique maximal ideal $\mathfrak{P}$ of a valuation ring $\mathfrak{O}$ of $\mathcal{K}_H$. The *residue field* of the place is $\mathfrak{O}/\mathfrak{P}$; it is a finite algebraic extension of L . A place $\mathfrak{P}$ *divides* X if $\mathfrak{P} \cap L[X]$ is equal to the ideal $XL[X]$.

There exists a mapping Φ from the set of parametrizations as above to the set of places of $\mathcal{K}_H$, defined by:

$$\Phi(R) = \{f(X, Y) \in \mathcal{K}_H \mid v_T(f(X(T), Y(T))) > 0\}.$$

The following definition was proposed by Duval [19]. We warn the reader that a slightly different definition appeared in [20] and [49], which corresponds to $\bar{L}$ -RPE in the sense below.

Definition 3 (*Rational Puiseux Expansions—RPEs*)

- If H is irreducible in $L[X, Y]$ and $H \neq Y$, a *system of L -RPEs above 0 of H* is a set of irreducible parametrizations $\{R_i\}_{1 \leq i \leq r}$ of the form:

$$R_i(T) = (X_i(T), Y_i(T)) = \left(\gamma_i T^{e_i}, \sum_{k=n_i}^{\infty} \beta_{ik} T^k \right) \in \bar{L}((T))^2$$

with $e_i > 0$, $n_i \in \mathbb{Z}$ and $\beta_{in_i} \neq 0$, and such that:

- (i) The mapping Φ above defines a one-to-one correspondence between the $\{R_i\}_{1 \leq i \leq r}$ and the places $\{\mathfrak{P}_i\}_{1 \leq i \leq r}$ of $\mathcal{K}_H$ dividing X ;
- (ii) The coefficient field of R_i is isomorphic to the residue field k_i of $\mathfrak{P}_i$.
- If $H = Y$, then $R_1 = (T, 0)$ is the only RPE³ of H above 0 and we set $n_1 = 0$. In this case, $e_1 = 1$ and $k_1 = L$.
- Assume that H is separable. A *system of L -RPEs above 0 of H* is the union of systems of L -RPEs for the irreducible factors of H in $L[X, Y]$.

The integer e_i is called the *ramification index* of R_i . In the sequel, we will identify k_i with the coefficient field of R_i and define $f_i = [k_i : L]$.

Definition 4 We say that R_i is *defined at $T = 0$* if $Y_i \in \bar{L}[[T]]$. In this case, the *center* of R_i is the pair $(X_i(0), Y_i(0)) \in k_i^2$.

Since L and H satisfy the characteristic condition (1), a system of L -RPEs exists. It is not unique since T may be replaced in R_i by γT for any γ in k_i .

The following result is classical [8, Chapter 4, Sect. 1]:

Theorem 1 $\sum_{i=1}^r e_i f_i = \deg_Y(H)$.

It is well-known [48] that the roots of H , viewed as a univariate polynomial in Y , may be expressed as fractional power series in X :

Theorem 2 (Puiseux) *There exist positive integers $e_1, \dots, e_s$ with $\sum_{i=1}^s e_i = \deg_Y(H)$ and $\deg_Y(H)$ distinct series:*

$$S_{ij}(X) = \sum_{k=n_i}^{\infty} \alpha_{ik} \zeta_{e_i}^{jk} X^{\frac{k}{e_i}}$$

where $1 \leq i \leq s$, $0 \leq j \leq e_i - 1$, $n_i \in \mathbb{Z}$ and $\alpha_{in_i} \neq 0$ if $S_{ij} \neq 0$, such that:

³ This case could have been avoided, but our symbolic-numeric method may cause an expansion to vanish modulo a prime p and we have preferred to provide a treatment for vanishing roots as well.

$$H(X, S_{ij}(X)) = 0 \text{ in } \overline{L} \left((X^{1/e_i}) \right).$$

If $S_{ij} = 0$ is a root of H , we set $n_i = 0$ and $e_i = 1$.

Definition 5 The S_{ij} are the *classical Puiseux series (CPS)* of H above 0.

From a system of RPEs, CPS can readily be computed with the following process:

1. R_i has exactly f_i conjugates over L , that we denote R_i^σ ($1 \leq \sigma \leq f_i$):

$$R_i^\sigma(T) = (X_i^\sigma(T), Y_i^\sigma(T)) = \left(\gamma_i^\sigma T^{e_i}, \sum_{k=n_i}^{\infty} \beta_{ik}^\sigma T^k \right).$$

2. Each R_i^σ yields a Puiseux series $S_i = Y_i^\sigma \left((X/\gamma_i^\sigma)^{1/e_i} \right)$.
3. The group $\mathbb{G}_{e_i}$ acts on S_i ; this gives the $\deg_Y(H)$ CPS of H .

Definition 6 Define $s_i = \min \{0, n_i\}$. The *regularity index* r_{ij} of S_{ij} in H is the least integer $N \geq s_i$ such that $\lceil S_{ij} \rceil_{e_i}^N = \lceil S_{uv} \rceil_{e_i}^N$ implies $(u, v) = (i, j)$. The truncated series $\lceil S_{ij} \rceil_{e_i}^{r_{ij}}$ is called the *singular part* of S_{ij} in H .

Roughly speaking, the regularity index is the number of terms necessary to “separate” a CPS from all the others. Since regularity indices of all Puiseux series corresponding to the same RPE are equal, we define the *singular part* of an RPE R_i to be the pair:

$$\left(\gamma_i T^{e_i}, \sum_{k=s_i}^{r_i} \beta_{ik} T^k \right)$$

where r_i is the regularity index of any Puiseux series associated to R_i .

We insist on the dependence on H of the regularity index: Indeed, the regularity index depends not only on the root considered, but also on the the initial terms of the other roots. For instance, the regularity index of the root 0 in $H = Y(Y - X)$ (resp. $H = Y(Y - X^3)$, $H = Y(X^3Y - 1)$) is 1 (resp. 3, 0). The regularity indices of the second roots are respectively 1, 3 and -3. Observe also that the singular part of a non zero root may be null: The regularity index of the root X^2 in $H = (Y - X^2)(Y - X)$ is 1 and the corresponding singular part is $0 + 0X = 0$. But the regularity index of the very same root in $H = (Y - X^2)(Y - X^3)$ is 2, yielding the singular part X^2 . Finally, consider the example $H = (Y - 1 - X)(X^2Y - 1)$. The regularity index of $1 + X$ is 0 and the regularity index of $1/X^2$ is -2; the singular parts are 1 and $1/X^2$. The above examples justify the introduction of the quantity s_i , that will also enter our complexity estimates.

Our goal is to compute singular parts of RPEs, since higher order terms of the series may be computed by means of asymptotically fast Newton iterations; see [26] or [46, Chapter 9].

3 Rational Newton-Puiseux algorithm

In order to compute singular parts of RPEs of H above 0, we present a modified version of Duval's rational Newton-Puiseux algorithm. In our approach, classical Newton polygons are replaced with *generic Newton polygons*. Those were introduced in the context of our symbolic-numeric algorithm; see [32, 35]. Unlike classical Newton polygons, they also allow to compute exactly regularity indices, even when one of the expansion is null.

Assume that $H = \sum_{i,j} a_{ij} X^j Y^i$ satisfies the same hypotheses as in the previous section; in particular, $H(0, Y) \neq 0$ and we define $\mathcal{I}(H) = v_Y(H(0, Y))$.

Definition 7 For each pair (i, j) of $\text{Supp}(H) = \{(i, j) \in \mathbb{N}^2 \mid a_{ij} \neq 0\}$, define $Q_{ij} = \{(i', j') \in \mathbb{R}^2 \mid i' \geq i \text{ and } j' \geq j\}$. Then the *Newton Polygon* $\mathcal{N}(H)$ of H is the set of finite edges of the convex hull $\mathcal{H}$ of $Q(H) = \cup_{(i,j) \in \text{Supp}(H)} Q_{ij}$. The *generic Newton polygon* $\mathcal{GN}(H)$ is obtained by restricting $\mathcal{N}(H)$ to edges with slope no less than -1 and by joining the leftmost remaining point to the vertical axis with an edge of slope -1 . Namely, we add a fictitious point $(0, j_0)$ to $\text{Supp}(H)$ so as to mask edges with slope less than -1 .

Examples and figures are given in [32, 35]. The algorithm first stage requires a special treatment and we introduce *exceptional polygons* for this purpose.

Definition 8 The exceptional Newton polygon $\mathcal{EN}(H)$ is the lower part of the convex hull of $\text{Supp}(H) \cup \{(0, 0)\}$.

In other words, it consists of the edge $[(0, 0), (\mathcal{I}(H), 0)]$, followed by a sequence of edges with positive slopes that join $(\mathcal{I}(H), 0)$ to $(\deg_Y(H), v_X(\text{lc}_Y(H)))$. In particular, $\mathcal{EN}(H) = [(0, 0), (\deg_Y(H), 0)]$ if H is monic.

Each edge Δ of $\mathcal{GN}(H)$ (resp. $\mathcal{N}(H)$, $\mathcal{EN}(H)$) corresponds to three integers q, m and l with $q > 0$, q and m coprime, such that Δ is on the line $qj + mi = l$. If Δ is the horizontal edge of $\mathcal{EN}(H)$, $m = l = 0$ and we choose $q = 1$.

Definition 9 We define the *characteristic polynomial* ϕ_Δ :

$$\phi_\Delta(T) = \sum_{(i,j) \in \Delta} a_{ij} T^{\frac{i-i_0}{q}}$$

where i_0 is the smallest value of i such that (i, j) belongs to Δ .

The rational Newton-Puiseux algorithm below performs successive changes of variable, determined by (q, m, l) and the roots of ϕ_Δ . It returns a set of triplets $\{(G_i(X, Y), P_i(X), Q_i(X, Y))\}_i$ such that:

- $G_i \in \overline{L}[X, Y]$,
- $P_i(X)$ is a monomial of the form $\lambda_i X^{e_i}$, $\lambda_i \in \overline{L}$,
- $Q_i(X, Y) = Q_{i0}(X) + c_i Y X^{r_i}$, where $(P_i(T), Q_{i0}(T))$ is the singular part of an RPE of F , r_i its regularity index, and $c_i \in \overline{L}$.

- $G_i(0, 0) = 0$, $G_{iY}(0, 0) \neq 0$ and $G_i(X, Y) = F(P_i(X), Q_i(X, Y))/X^{L_i}$, for some non negative integer L_i .

Let k_i be the field generated over L by the coefficients of P_i and Q_i . By the formal Implicit Function Theorem, the latter conditions ensure that there exists a unique power series $U_i \in k_i[[X]]$ such that $G_i(X, U_i(X)) = 0$ and $U_i(0) = 0$. The corresponding parametrization of F is therefore $R_i(T) = (P_i(T), Q_i(T, U_i(T)))$. Hence, we will consider that a triplet (G_i, P_i, Q_i) represents an RPE with residue field k_i .

We first give specifications for two sub-algorithms:

Factor(L, ϕ)

Input : L , a field, and ϕ , a univariate polynomial in $L[T]$.

Output : A set $\{(\phi_i, M_i)\}_i$ with ϕ_i irreducible in $L[T]$, $c \in L$ and $\phi = c \prod_i \phi_i^{M_i}$.

Bézout(q, m)

Input : $(q, m) \in \mathbb{Z}^2$ with $q > 0$.

Output : $(a, b) \in \mathbb{Z}^2$ such that $aq - bm = 1$ and $0 \leq b < q$.

The rational Newton-Puiseux algorithm below (see Fig. 1) is presented in a recursive setting and we suppose that a mechanism to distinguish initial calls from recursive calls exists; for instance, an additional parameter could be used. Note that the main transformation is performed modulo a power of X . We will discuss in Sect. 5 how this parameter should initially be chosen and updated.

Example 1 Using a balanced representation for $\mathbb{F}_{13}$, consider $F \in \mathbb{F}_{13}[X, Y]$:

$$F(X, Y) = (3X^3 - Y^2 + 2Y - 1) \left(-2X^2 + Y^2 - 2Y + 1 \right) (XY^3 - 2).$$

The output of the algorithm is:

$$\begin{aligned} (P_1, Q_1) &= (-4X^2, 1 + X^2(0 - 4X(1 + Y))) \\ (P_2, Q_2) &= (X, 1 + X(\sqrt{2} + Y)) \\ (P_3, Q_3) &= \left(2X^3, \frac{1}{X}(1 + Y) \right) \end{aligned}$$

where $\sqrt{2}$ stands for any root of the irreducible polynomial $T^2 - 2 \in \mathbb{F}_{13}[T]$. Null coefficients correspond to fictitious edges of generic polygons. Ramifications indices are respectively 2, 1 and 3. Regularity indices are respectively 3, 1 and -1 . Residue fields are $\mathbb{F}_{13}$, $\mathbb{F}_{13}(\sqrt{2})$ and $\mathbb{F}_{13}$.

Generic Newton polygons and multiplicity structures for characteristic polynomials encountered in the execution of the algorithm form *polygon trees*; see [33, Section 3.3] or [32, Section 2.1.5] for a formal definition. Polygon trees encode the symbolic information needed to guide floating point computations [31, 32].

RNPuiseux(L, H, N)

Input:

L : A field.

H : A separable element of $L[X, Y]$, with $\deg_Y(H) \geq 1$ and $H(0, Y) \neq 0$.

N : A positive integer (truncation order).

Output:

A set of triplets $\{[G_i, P_i, Q_i]\}_i$, which form a set of representatives for:

- L -RPEs of H above 0 for the initial call,
- L -RPEs of H centered at $(0, 0)$ for recursive calls.

Begin

If in a recursive call then

$\mathcal{N} \leftarrow \mathcal{GN}(H)$

If $\mathcal{I}(H) = 1$ then Return $\{[H, X, Y]\}$ End

$\tilde{H} \leftarrow H$

else

$\mathcal{N} \leftarrow \mathcal{EN}(H)$

$v \leftarrow v_X(\text{lc}_Y(H))$

$\tilde{H} \leftarrow H \bmod X^{N+v+1}$

End

$\mathcal{R} \leftarrow \{\}$

For each side Δ of $\mathcal{N}$ do

Compute q, m, l and ϕ_Δ

$(a, b) \leftarrow \text{Bézout}(q, m)$

For each (f, k) in $\text{Factor}(L, \phi_\Delta)$ do

$\xi \leftarrow \text{Any root of } f$

$\hat{N} \leftarrow \lfloor N/[L(\xi) : L] \rfloor$ // Update truncation order

$\hat{H}(X, Y) \leftarrow \tilde{H}(\xi^b X^q, X^m(\xi^a + Y))/X^l \bmod X^{\hat{N}+1}$

For each $[G, P, Q]$ in RNPuiseux($L(\xi), \hat{H}, \hat{N}$) do

$\mathcal{R} \leftarrow \mathcal{R} \cup \{[G, \xi^b P^q, P^m(\xi^a + Q)]\}$

End

End

End

Return $\mathcal{R}$

End.

Fig. 1 The rational Newton-Puiseux algorithm

The separability hypothesis for H reduces to squarefreeness because of our characteristic condition (1). It can easily be achieved using Yun's algorithm [51] or variants if $p > 0$; see [28] for recent results.

4 Truncation order

We now deal with parameter N of RNPuiseux. We first introduce notations and make some assumptions:

- We write $F(X, Y) = \sum_{i=0}^{d_Y} A_i(X)Y^i$.
- We assume that the ground field L satisfies characteristic condition (1).
- $\{R_i\}_{1 \leq i \leq \rho}$ with $R_i(T) = (X_i(T), Y_i(T))$ stands for singular parts of a system of RPEs above 0, with coefficient fields k_i .

- (G_i, P_i, Q_i) is the output of `RNPuiseux` corresponding to R_i .
- $(n_i, s_i, r_i, e_i, f_i)$, $1 \leq i \leq \rho$ are the integers associated with R_i in Sect. 2.
- The CPS associated with R_i are denoted S_{ijk} , $0 \leq k \leq e_i - 1$, $1 \leq j \leq f_i$.
- We define:

$$\delta_F = \sum_{i=1}^{\rho} (r_i - s_i) f_i \quad \eta_F = \sum_{i=1}^{\rho} (r_i - s_i + 1) f_i.$$

In particular, if F is monic, all the s_i are null and the definition of δ_F is identical to that of [32, 33]. This modification allows to deal with the non monic case. Considering Theorem 1, remark that:

$$0 \leq \delta_F \leq \eta_F = \delta_F + \sum_{i=1}^{\rho} f_i \leq \delta_F + d_Y.$$

Note also, that unlike δ_F , η_F cannot be null.

Remark 1 The integer η_F is exactly the number of elements of L necessary to represent the Y_i . If $n_i \geq 0$, assume that a vector of coefficients in k_i , indexed from 0 to r_i , is used to represent the singular part of $Y_i(T)$. Such a representation is legitimate since generic and exceptional Newton polygons may introduce null coefficients. If $n_i < 0$, then Y_i may be represented by a vector of $r_i - n_i + 1$ coefficient in k_i . Since the elements of k_i may in turn be represented by vectors of f_i elements of L , the total size of `RNPuiseux` output is essentially η_F .

In this section, we first show that δ_F may be used as an order of truncation, then we relate δ_F to R_F .

4.1 Truncating powers of X

Lemma 1 Define $v = v_X(A_{d_Y})$. Then, $v = -\sum_{i=1}^{\rho} s_i f_i$.

Proof The minimal polynomial of S_{ijk} over $\bar{L}((X))$ is $M_{ij} = \prod_{k=0}^{e_i-1} (Y - S_{ijk})$. The content of M_{ij} with respect to Y is X^{s_i} . Setting $M = \prod_{ij} M_{ij}$, it is easily seen that M and $X^{-v} F$ are equal, up to a factor that is a unit in $\bar{L}[[X]]$. By Gauss Lemma, the content of M is $X^{\sum_{i=1}^{\rho} s_i f_i}$, while the content of $X^{-v} F$ is X^{-v} .

Proposition 1 RPEs of $\lceil F \rceil^{\delta_F+v}$ and F above 0 have the same singular parts.

Proof Since F and $\lceil F \rceil^{\delta_F+v}$ have the same degree in Y , they have the same number of CPSs above 0; those of $\lceil F \rceil^{\delta_F+v}$ are denoted by $\{U_m\}_{1 \leq m \leq d_Y}$. We order the R_i so that:

$$\frac{r_1 - s_1}{e_1} \leq \frac{r_2 - s_2}{e_2} \leq \dots \leq \frac{r_{\rho} - s_{\rho}}{e_{\rho}}. \quad (2)$$

It is sufficient to show that for each triplet (i, j, k) there exists an integer $l(i, j, k)$ such that $v_X(S_{ijk} - U_{l(i,j,k)}) > r_i/e_i$. In other words, $U_{l(i,j,k)} = \lceil S_{ijk} \rceil^{\frac{r_i}{e_i}} + \text{higher order terms}$. This will prove in particular that $\lceil F \rceil^{\delta_F+v}$ is squarefree and that l defines a one-to-one map.

Assume that Proposition 1 is wrong. Let i_0 , $1 \leq i_0 \leq \rho$, be the smallest integer such that there exist j_0 and k_0 satisfying: For each m , $v_X(S_{i_0 j_0 k_0} - U_m) \leq r_{i_0}/e_{i_0}$.

Set $m_0 = \sum_{i=1}^{i_0-1} e_i f_i$. For $i < i_0$ and all relevant j and k , there exists pairwise distinct $l(i, j, k)$ with $v_X(S_{ijk} - U_{l(i,j,k)}) > r_i/e_i$; we assume that the U_m are ordered so that $l(i, j, k) \leq m_0$.

Write $F = \lceil F \rceil^{\delta_F+v} + X^{\delta_F+v+1}V$, where $V \in L[X, Y]$. Computing valuations in $\lceil F \rceil^{\delta_F+v}(X, S_{i_0 j_0 k_0}) = -X^{\delta_F+v+1}V(X, S_{i_0 j_0 k_0})$, we obtain:

$$v + \sum_{m=1}^{d_Y} v_X(S_{i_0 j_0 k_0} - U_m) = v + \delta_F + 1 + v_X(V(X, S_{i_0 j_0 k_0})). \quad (3)$$

Since $v_X(V(X, S_{i_0 j_0 k_0})) \geq d_Y \frac{s_{i_0}}{e_{i_0}}$, using the definition of i_0 , we get:

$$d_Y \frac{s_{i_0}}{e_{i_0}} + \delta_F + 1 \leq \sum_{m=1}^{m_0} v_X(S_{i_0 j_0 k_0} - U_m) + (d_Y - m_0) \frac{r_{i_0}}{e_{i_0}}.$$

Assumption (2) implies $\sum_{i=i_0}^{\rho} (r_i - s_i) f_i \geq (d_Y - m_0)(r_{i_0} - s_{i_0})/e_{i_0}$. Hence:

$$\sum_{i=1}^{i_0-1} (r_i - s_i) f_i + 1 \leq \sum_{m=1}^{m_0} \left(v_X(S_{i_0 j_0 k_0} - U_m) - \frac{s_{i_0}}{e_{i_0}} \right). \quad (4)$$

Next, we denote $I_<$ (resp. $I_>$, $I_=\$) the subset of integers in $[1, i_0 - 1]$ such that $\frac{s_i}{e_i} < \frac{s_{i_0}}{e_{i_0}}$ (resp. $\frac{s_i}{e_i} > \frac{s_{i_0}}{e_{i_0}}$, $\frac{s_i}{e_i} = \frac{s_{i_0}}{e_{i_0}}$). For $i < i_0$ and all relevant j and k , we have:

- If $i \in I_=\$, then $v_X(S_{i_0 j_0 k_0} - U_{l(i,j,k)}) \leq r_i/e_i$; otherwise, $S_{i_0 j_0 k_0}$ and S_{ijk} would coincide up to an order greater than r_i/e_i , contradicting the definition of r_i .
- If $i \in I_<$, then $v_X(S_{i_0 j_0 k_0} - U_{l(i,j,k)}) = s_i/e_i$.
- If $i \in I_>$, then $v_X(S_{i_0 j_0 k_0} - U_{l(i,j,k)}) = s_{i_0}/e_{i_0}$.

Splitting the right hand sum in inequality (4) and using this remark, we obtain:

$$\sum_{i=1}^{i_0-1} (r_i - s_i) f_i + 1 \leq \sum_{i \in I_<} \left(\frac{s_i}{e_i} - \frac{s_{i_0}}{e_{i_0}} \right) e_i f_i + \sum_{i \in I_=} \left(\frac{r_i}{e_i} - \frac{s_i}{e_i} \right) e_i f_i.$$

But in the right hand side, the first sum is negative. Hence:

$$\sum_{i=1}^{i_0-1} (r_i - s_i) f_i + 1 \leq \sum_{i \in I_=} (r_i - s_i) f_i.$$

This is clearly impossible since all terms are non negative and $I_{\infty} \subset [1, i_0 - 1]$; the proof of Proposition 1 is complete.

The initial term of an RPE centered at $(0,0)$ corresponds to an edge with negative slope of the classical Newton polygon of F . We assume for a moment that the $\{R_i\}_{1 \leq i \leq \rho'}$, $1 \leq \rho' \leq \rho$, are exactly the RPEs of F centered at $(0,0)$ and we introduce the following quantity:

$$\theta_F = \sum_{i=1}^{\rho'} (r_i - s_i) f_i = \sum_{i=1}^{\rho'} r_i f_i.$$

We recall that S_{ijk} vanishes at $X = 0$ if $1 \leq i \leq \rho'$.

Proposition 2 *RPEs centered at $(0, 0)$ of $\lceil F \rceil^{\theta_F}$ and F have the same singular parts.*

Proof Although this proof is similar to that of Proposition 1, we have not found a reduction to Proposition 1, or vice versa. Define $d'_Y = \deg_Y(\lceil F \rceil^{\theta_F}) \leq d_Y$, $v' = v_X(\text{lc}_Y(\lceil F \rceil^{\theta_F}))$ and:

$$d_0 = v_Y(F(0, Y)) = v_Y(\lceil F \rceil^{\theta_F}(0, Y)) = \sum_{i=1}^{\rho'} e_i f_i$$

Here, d_0 represents the number of CPSs of F and $\lceil F \rceil^{\theta_F}$ vanishing at $X = 0$. We assume $d_0 > 0$, otherwise, there is nothing to prove. Denote by $\{U_m\}_{1 \leq m \leq d'_Y}$ the Puiseux series of $\lceil F \rceil^{\theta_F}$ and suppose that they are ordered so that U_m vanishes at $X = 0$ for $1 \leq m \leq d_0$. Moreover, without loss of generality, we assume:

$$r_1/e_1 \leq r_2/e_2 \leq \dots \leq r_{\rho'}/e_{\rho'}. \quad (5)$$

If, for each triplet (i, j, k) with $1 \leq i \leq \rho'$, there exists an integer $l(i, j, k)$ with $1 \leq l(i, j, k) \leq d_0$ and $v_X(S_{ijk} - U_{l(i, j, k)}) > r_i/e_i$, we are done. Suppose that i_0 is the smallest integer such that there exist j_0 and k_0 satisfying: $v_X(S_{i_0 j_0 k_0} - U_m) \leq r_{i_0}/e_{i_0}$ for all m , $1 \leq m \leq d_0$. Set $m_0 = \sum_{i=1}^{i_0-1} e_i f_i$. For $i < i_0$ and all relevant j and k , there exists pairwise distinct $l(i, j, k)$ with $v_X(S_{ijk} - U_{l(i, j, k)}) > r_i/e_i$; we assume that the U_m are ordered so that $l(i, j, k) \leq m_0$ for $i < i_0$.

Evaluating the equality $F = \lceil F \rceil^{\theta_F} + X^{\theta_F+1} V$ at $Y = S_{i_0 j_0 k_0}$, we get:

$$\theta_F + 1 + v_X(V(X, S_{i_0 j_0 k_0})) = v' + \sum_{m=1}^{d'_Y} v_X(S_{i_0 j_0 k_0} - U_m).$$

For $m > d_0$, $v_X(S_{i_0 j_0 k_0} - U_m) = v_X(U_m) \leq 0$ and Lemma 1 applied to $\lceil F \rceil^\theta$ gives $v' = -\sum_{m=d_0+1}^{d'_Y} v_X(U_m)$. Since $v_X(V(X, S_{i_0 j_0 k_0})) \geq 0$, we get:

$$\theta_F + 1 \leq \sum_{m=1}^{d_0} v_X(S_{i_0 j_0 k_0} - U_m).$$

We have $v_X(S_{i_0 j_0 k_0} - U_m) \leq r_i/e_i$ if $m = l(i, j, k) \leq m_0$ for some $i < i_0$ and $v_X(S_{i_0 j_0 k_0} - U_m) \leq r_{i_0}/e_{i_0}$ for $m > m_0$. We deduce:

$$\theta_F + 1 \leq \sum_{i=1}^{i_0-1} r_i f_i + (d_0 - m_0) \frac{r_{i_0}}{e_{i_0}} \iff \sum_{i=i_0}^{\rho'} r_i f_i + 1 \leq (d_0 - m_0) \frac{r_{i_0}}{e_{i_0}}.$$

The latter inequality is impossible because of (5); Proposition 2 is proved.

Consider the following examples:

- If $F_m = X^{2m} Y^2 - 1$, the two CPS are $\pm 1/X^m$. We have $\delta_{F_m} = 2(-m + m) = 0$, but $\delta_{F_m} + v = 2m$. The bound $\delta_{F_m} + v$ is optimal for this case.
- Define $F_m = (Y - X^m)(X^m Y - 1)$ for $m > 0$. We have $\theta_{F_m} = 0$; $\lceil F_m \rceil^0 = Y$ gives the correct singular part for the positive order expansion, namely, 0. But to get both singular parts, order $\delta_{F_m} + v = 0 + (-m - (-m)) + m = m$ is required.
- For $F_{e,r} = Y^e - X^r$ with e and r coprime, we have $\theta_{F_{e,r}} = r$, but $\lceil F_{e,r} \rceil^{r-1} = Y^e$ is not even squarefree.
- Let Z_{ijk} be the singular parts of CPS of some polynomial F and set $H = X^v \prod_{ijk} (Y - Z_{ijk}) \in L[X, Y]$. We define $\gamma_F = \sum_{i=1}^{\rho} (\max\{r_i, 0\} - s_i) f_i$. It can be shown that $\deg_X(H) = \gamma_F \leq \delta_F + v$. Truncating at order γ_F is not always sufficient, as demonstrated by the following example:

$$F = (X^3 + 9X^4) Y^6 - 3Y^4 X^2 + (3X - 6X^2) Y^2 - X^2 - 2X - 1.$$

The singular part of the unique RPE is $(T^6, \frac{1}{T^3} - \frac{1}{2} \frac{1}{T})$ and $\gamma_F = 0 - (-3) = 3$.

But the RPE of $\lceil F \rceil^3$ is precisely $(T^6, \frac{1}{T^3} + \frac{1}{T})$.

Although bounds of Propositions 1 and 2 are attained for families of examples, they are usually not optimal.

For the next step, we introduce more notations. If H is a polynomial as specified in RNPuiseux, let Δ be an edge of the Newton polygon, ξ be a root of ϕ_Δ with multiplicity $M_{\Delta, \xi}$ and $(m_\Delta, q_\Delta, l_\Delta, a_\Delta, b_\Delta)$ be the integers associated with Δ . The main transformation in the rational Newton-Puiseux algorithm is the computation of $H_{\Delta, \xi}$:

$$H_{\Delta, \xi} = \frac{(\xi^{b_\Delta} X^{q_\Delta}, X^{m_\Delta} (\xi^{a_\Delta} + Y))}{X^{l_\Delta}}.$$

Proposition 3

$$\theta_H = \sum_{(\Delta, \xi) \in \mathcal{GN}(H)} [L(\xi) : L] (\theta_{H_{\Delta, \xi}} + m_{\Delta} M_{\Delta, \xi}) \quad (6)$$

$$\delta_H = \sum_{(\Delta, \xi) \in \mathcal{EN}(H)} [L(\xi) : L] \theta_{H_{\Delta, \xi}}. \quad (7)$$

Proof Let R_i be an RPE of H and $\widehat{R}_i$ be the corresponding RPE of $H_{\Delta, \xi}$. If r_i and e_i (resp. $\widehat{r}_i$ and $\widehat{e}_i$) are the regularity and ramification indices of R_i (resp. $\widehat{R}_i$), we have: $\widehat{r}_i = r_i - m_{\Delta}/q_{\Delta} e_i = r_i - m_{\Delta} \widehat{e}_i$. Moreover, if f_i (resp. $\widehat{f}_i$) is the degree of the residue field of R_i (resp. $\widehat{R}_i$), then $f_i = [L(\xi) : L] \widehat{f}_i$ because the ground field of $H_{\Delta, \xi}$ is $L(\xi)$. Denote by $E_{\Delta, \xi}$ the set of indices i such that R_i corresponds to (Δ, ξ) . Then:

$$\begin{aligned} \theta_H &= \sum_{(\Delta, \xi) \in \mathcal{GN}(H)} \sum_{i \in E_{\Delta, \xi}} r_i f_i \\ &= \sum_{(\Delta, \xi) \in \mathcal{GN}(H)} \sum_{i \in E_{\Delta, \xi}} (\widehat{r}_i + m_{\Delta} \widehat{e}_i) \widehat{f}_i [L(\xi) : L] \\ &= \sum_{(\Delta, \xi) \in \mathcal{GN}(H)} [L(\xi) : L] \left(\sum_{i \in E_{\Delta, \xi}} \widehat{r}_i \widehat{f}_i + m_{\Delta} \sum_{i \in E_{\Delta, \xi}} \widehat{e}_i \widehat{f}_i \right). \end{aligned}$$

The latter sum represents the number of roots of $H_{\Delta, \xi}$ that vanish at $X = 0$, and this is precisely the multiplicity of ξ as a root of Φ_{Δ} ; see [6] Section 8.3 or [48] for instance. This gives equality (6). For the second equality, the same argument applied to the exceptional Newton polygon gives:

$$\sum_{i=1}^{\rho} r_i f_i = \sum_{(\Delta, \xi) \in \mathcal{EN}(H)} [L(\xi) : L] (\theta_{H_{\Delta, \xi}} + m_{\Delta} M_{\Delta, \xi}).$$

For $\Delta \in \mathcal{EN}(H)$, we always have $m_{\Delta} \leq 0$. From $\frac{m_{\Delta}}{q_{\Delta}} = \frac{s_i}{e_i}$, we get:

$$\sum_{(\Delta, \xi) \in \mathcal{EN}(H)} [L(\xi) : L] m_{\Delta} M_{\Delta, \xi} = \sum_{(\Delta, \xi) \in \mathcal{EN}(H)} \frac{m_{\Delta}}{q_{\Delta}} \sum_{i \in E_{\Delta, \xi}} e_i f_i = \sum_{i=1}^{\rho} s_i f_i.$$

Proposition 3 is proved.

The main result of this section is now:

Proposition 4 *The function call $\text{RNpuiSeux}(L, F, \delta_F)$ returns a system of L -RPEs of F above 0.*

Proof We consider a single expansion R and drop indices to simplify notations, so that (G, P, Q) are the associated quantities. Let $F = H_1, H_2, \dots, H_{h+1} = G$ be

the sequence of input polynomials in the corresponding branch of the function call tree. Propositions 1 and 2 show that it is sufficient to truncate F modulo X^{δ_F+v+1} and to compute H_i modulo $X^{\theta_{H_i}+1}$ for $i > 1$. But coefficients of the recurrence relation in Proposition 3 are positive; therefore, $\delta_F/[L(\xi_1, \dots, \xi_i) : L] \geq \theta_{H_{i+1}}$ for $1 \leq i \leq h$, where the ξ_i are the successive roots of the characteristic polynomials. Hence, `RNPuiseux` returns the expected output.

In practice, δ_F is usually unknown in advance and we will give bounds in Sect. 4.2.

Remark 2 Proposition 3 allows to truncate further powers of X in `RNPuiseux`. Consider the following example: Assume that L is a field that contains no square roots of 2 and let F be the irreducible polynomial of $L[X, Y]$ corresponding to the RPE $(T^6, T^2 + \sqrt{2}T^4 + T^9)$, with residue field of degree 2 over L . We skip the first step, because the exceptional polygon has a unique slope with a unique root 0. Since F is monic, $\delta_F + v = \theta_{H_1}$. Each polygon has a unique slope that we call Δ_i . With notations of the proof of Proposition 4, starting from $\theta_{H_1} = 2 \times 9 + 0 = 18$ and applying relation (6), we obtain $\theta_{H_2} = \theta_{H_1} - M_{\Delta_1, 1} = 18 - 4 = 14$, $\theta_{H_3} = \theta_{H_2} / \left[L(\sqrt{2}) : L \right] - M_{\Delta_2, \sqrt{2}} = 14/2 - 2 = 5$ (it can be shown that the optimal truncation orders are 7, 9 and 5). Hence, if there is a unique RPE, the θ_{H_i} , $i > 1$, can be deduced from $\delta_F + v$. If there are several RPEs, information obtained from recursive calls for some branches can also be used to reduce truncation orders for other branches.

Remark 3 The proof given in [32, 34] for Proposition 4 is significantly different: Therein, a formula for the optimal truncation order is determined and it is shown that it is bounded by δ_F (F is assumed monic). Note that the proof of Lemma 16 of [32] is wrong: It is implicitly assumed that the sequence of optimal truncation orders is decreasing, but this is false; see the example of Remark 2. However, the final result is correct because one can show that the optimal truncation orders are bounded by terms of the decreasing sequence $\delta_F + v \geq \theta_{H_2} \geq \dots \geq \theta_{H_h}$, as above.

To conclude this section, we show that there are families of examples such that the sequence of truncation orders given by Propositions 1 and 2 is optimal: This occurs when the ramification is introduced at the last step of the computation. Consider the irreducible polynomial F that vanishes at $(T^6, 1 + T^6 + T^{12} + T^{13})$. There is a unique RPE above 0 and $\mathcal{EN}(F)$ has a unique horizontal slope. The unique root of the characteristic polynomial is 1. Therefore, $\delta_F + v = \theta_{H_1} = \theta_{H_2} = 13$, $\theta_{H_3} = 7$, $\theta_{H_4} = 1$. Truncating the H_i at orders less than 13, 7 or 1 yields an incorrect output: Indeed, $\lceil F \rceil^{12}$, $\lceil H_2 \rceil^{12}$, $\lceil H_3 \rceil^6$ and $\lceil H_4 \rceil^0$ are not even squarefree.

4.2 Bounding δ_F

The quantities δ_F and η_F are a priori unknown. To obtain a complete algorithm, we provide a bound for δ_F .

Proposition 5 *We have: $\delta_F + v \leq v_X(R_F)$.*

Proof If F has only one Puiseux series above 0, $\delta_F = 0$, $R_F = \text{lc}_Y(F)$ and the result holds. Assume that there exist at least two Puiseux series and let S_{ijk} be such a series. We first prove the following inequality:

$$\sum_{\substack{(i', j', k') \\ (i', j', k') \neq (i, j, k)}} v_X(S_{ijk} - S_{i'j'k'}) - \min \left\{ \frac{s_i}{e_i}, \frac{s_{i'}}{e_{i'}} \right\} \geq \frac{r_i - s_i}{e_i}. \quad (8)$$

Since the left hand side terms are non negative, it is enough to prove the inequality for one term of the sum, or for any sum of such terms. For any integer n , we denote $S_{ijk}^{[n, e_i]}$ the series obtained by applying the element $X \mapsto \zeta_{e_i}^n X^{1/e_i}$ of $\mathbb{G}_{e_i}$ to S_{ijk} .

If $r_i = s_i$, then (8) obviously holds. Assume in the sequel that $r_i > s_i$. Let $F = H_1, H_2, \dots, H_{h+1} = G_i$ be the sequence of input polynomials for RNPuiseux in the branch of the function call tree corresponding to S_{ijk} . Note that $r_i > s_i$ implies $h > 1$. Denote by $\mathcal{N}$ the generic Newton polygon of H_h , and by Δ and ξ the edge of $\mathcal{N}$ and the root of ϕ_Δ used in the transformation from H_h to H_{h+1} . Since $\mathcal{I}(H_{h+1}) = 1$, ξ has multiplicity 1 in ϕ_Δ ; see [6] Section 8.3 or [48]. The edge Δ has a slope $-m/q \in [-1, 0[$, where m and q are coprime integers, q divides e_i and $1 \leq m < q$. From [35, Section 4.4, Eq. 8 and Remark 33], there exists a non zero $\gamma \in \bar{L}$ such that:

$$S_{ijk} = \lceil S_{ijk} \rceil_{e_i}^{\frac{r_i-1}{e_i}} + \xi^{1/q} \gamma X^{\frac{r_i}{e_i}} + \dots$$

Three cases are possible:

- Case 1:** ϕ_Δ has a root $\xi_0 \neq \xi$. This root yields a CPS $S_{i_0j_0k_0} = \lceil S_{ijk} \rceil_{e_i}^{\frac{r_i-1}{e_i}} + \xi_0^{1/q} \gamma X^{\frac{r_i}{e_i}} + \dots$. Thus, $v_X(S_{ijk} - S_{i_0j_0k_0}) = r_i/e_i$ and (8) is true.
- Case 2:** $q > 1$. There exists a CPS $S_{ijk_0} = \lceil S_{ijk} \rceil_{e_i}^{\frac{r_i-1}{e_i}} + \zeta_q \xi^{1/q} \gamma X^{\frac{r_i}{e_i}} + \dots$. Hence $v_X(S_{ijk} - S_{ijk_0}) = r_i/e_i$ and (8) holds.
- Case 3:** ξ is the unique root of ϕ_Δ and $q = 1$. In particular, this means that the ramification index e_i was already computed after the $(h-1)$ th step of the algorithm, so that the polynomial H_{h-1} yielded the coefficient of $X^{\frac{r_i-1}{e_i}}$. Moreover, since $\mathcal{I}(H_h) > 1$, $\mathcal{N}$ has another edge Δ_0 , whose slope $-m_0/q_0$ must belong to $] -1, 0[$, with m_0 and q_0 coprime, and $1 \leq m_0 < q_0$. Let $S_{i_0j_0k_0}$ be a CPS corresponding to Δ_0 . There exists $\alpha_0 \neq 0$ such that, for all integers l , $0 \leq l < q_0$, we have:

$$S_{i_0j_0k_0}^{[l e_i, q_0 e_i]} = \lceil S_{ijk} \rceil_{e_i}^{\frac{r_i-1}{e_i}} + \alpha_0 \zeta_{q_0}^{m_0 l} X^{\frac{r_i-1}{e_i} + \frac{m_0}{q_0 e_i}} + \dots$$

Therefore:

$$\sum_{l=0}^{q_0-1} \left(v_X \left(S_{ijk} - S_{i_0j_0k_0}^{[l e_i, q_0 e_i]} \right) - \frac{s_i}{e_i} \right) = q_0 \frac{r_i - s_i}{e_i} + \frac{m_0 - q_0}{e_i}. \quad (9)$$

Remark that $q_0(r_i - s_i) + m_0 - q_0 = (r_i - s_i) + (q_0 - 1)(r_i - s_i - 1) + m_0 - 1$. But $r_i > s_i$ implies $r_i - s_i - 1 \geq 0$. As $q_0 > 1$ and $m_0 \geq 1$, the right hand side of (9) is no less than $(r_i - s_i)/e_i$ and (8) is proved in the three cases.

Finally, summing (8) over (i, j, k) , we obtain $v_X(R_F) - v(2d_Y - 1) - M \geq \delta_F$, where:

$$M = \sum_{(i,j,k)} \sum_{\substack{(i',j',k') \\ (i',j',k') \neq (i,j,k)}} \min \left\{ \frac{s_i}{e_i}, \frac{s_{i'}}{e_{i'}} \right\}.$$

To conclude the proof, we just need to show that $M \geq -v(2d_Y - 2)$. Since $s_i \leq 0$, we have $\min \left\{ \frac{s_i}{e_i}, \frac{s_{i'}}{e_{i'}} \right\} \geq \frac{s_i}{e_i} + \frac{s_{i'}}{e_{i'}}$ and:

$$M \geq 2 \sum_{(i,j,k)} \left(\sum_{(i',j',k')} \frac{s_{i'}}{e_{i'}} \right) - \frac{s_i}{e_i} = 2 \sum_{(i,j,k)} -v - \frac{s_i}{e_i} = -2v(d_Y - 1).$$

The last equalities come from Lemma 1.

5 Arithmetic complexity of RNPuiseux over finite fields

In this section, L denotes a finite field and $F = \sum_{i=0}^{d_Y} A_i(X)Y^i$ is a separable polynomial in $L[X, Y]$, primitive with respect to Y , with degrees $d_Y = \deg_Y(F)$, $d_X = \deg_X(F)$. We denote by $p > d_Y$ the characteristic of L and define $t_0 = [L : \mathbb{F}_p]$. The section is devoted to the proof of the following theorems:

Theorem 3 *The RNPuiseux algorithm can compute singular parts of a system of RPEs above 0 of F with an expected number of $O^{\sim}(d_Y^3 d_X^2 + d_Y^2 d_X t_0 \log p)$ field operations in L .*

As usual, the notation $O^{\sim}$ hides logarithmic factors. This result was presented in [33] without proof and only when F is a monic polynomial in Y . By considering all critical points of F , we get a remarkably similar result:

Theorem 4 *The RNPuiseux algorithm can be used to compute singular parts of systems of RPEs above all conjugacy classes over L of critical points of F , including the point at infinity, with an expected number of $O^{\sim}(d_Y^3 d_X^2 t_0 \log p)$ field operations in L .*

We immediately deduce:

Corollary 1 *Assume that F is irreducible in $\bar{L}[X, Y]$, with $p > d_Y$. There exists an algorithm to compute the genus of the curve $F(X, Y) = 0$ with an expected number of $O^{\sim}(d_Y^3 d_X^2 t_0 \log p)$ field operations in L .*

Proof Obvious, by Riemann-Hurwitz formula; see [21], for instance.

Some comments are necessary:

- Our computational model is very simple: following [46], we just count the number of arithmetic operations (addition, multiplication, division) in L . This is sufficient to provide running time upper estimates for any reasonable implementation with constant time access to polynomial coefficients. Most algorithms and subalgorithms are deterministic (RNPuiseux, changes of variable, fast polynomial multiplication, etc); for them, we consider the worst case. However, subalgorithms for factoring polynomials (Sect. 5.3) and computing primitive elements (Sect. 5.1) are probabilistic of Las Vegas type. Their running times depend on random choices of element in L ; hence, we use average running times, that propagate to our main results.
- To obtain Theorems 3 and 4, asymptotically fast multiplication of univariate polynomials over finite fields is required. See below.
- Theorem 3 improves results of Duval [20], who used the D5 approach [17] to avoid factorization. The author obtained $O(d_Y^6 d_X^2)$ operations in L for the monic case, and $O(d_Y^8 d_X^2)$ in general. If the cardinality of L is large, it may be preferable to use the D5 approach in order to remove the dependence on p and t_0 ; see [16] for complexity results.

Let us introduce some notations and recall a few facts:

- L_t denotes a *simple algebraic extension* of degree t of L (i.e. generated over L by a primitive element).
- By an L -operation, we mean a field operation in L : addition, multiplication or division.
- For a sufficiently large integer N , we define $\mathcal{L}(N) = \log N \log \log N$. In the sequel, $\log$ always stands for a logarithm with base greater than 1.
- $\mathcal{M}(N)$ is a bound for the number of field operations needed to compute the product of two polynomials of degree no larger than N with coefficients in a finite field. We will assume that $\mathcal{M}(N)$ is a non decreasing function with $\mathcal{M}(N) \geq N$ that satisfies the following properties:
 1. There exists a positive constant C such that, for any real numbers a and b greater than one and large enough $c = ab$, we have:

$$\mathcal{M}(a)\mathcal{M}(b) \leq C\mathcal{M}(c)\mathcal{L}(c). \quad (10)$$

2. There exists a positive constant C' such that, for any real numbers a and b greater than one:

$$\mathcal{M}(a) + \mathcal{M}(b) \leq C'\mathcal{M}(a + b). \quad (11)$$

For classical arithmetic, we have $\mathcal{M}(N) = C_0 N^2$ for some positive constant C_0 ; properties above are obviously satisfied. If asymptotically fast multiplication is used, we can choose $\mathcal{M}(N) = C_1 N \mathcal{L}(N) \in \mathcal{O}^\sim(N)$ for $N > N_0$ and $\mathcal{M}(N) = M_0$ for $N \leq N_0$, where N_0, M_0, C_1 are positive constants; see the seminal paper [7], see [39] for the most general setting, or [46, Theorem 8.23] for our context.

For $a, b > N_0$, note that $\log a \log b \leq \frac{1}{2} \log^2 ab$ implies $(\log \log a)(\log \log b) \leq 2(\log \log ab)^2$, and (10) follows. Other cases for (10) are obvious if $\mathcal{L}(c) \geq 1$, and so is (11).

- It will be convenient to introduce $\widetilde{\mathcal{M}}(N) = \mathcal{M}(N)/N$, so that $\widetilde{\mathcal{M}}(N) \in O(N)$ or $\widetilde{\mathcal{M}}(N) \in O(\mathcal{L}(N))$.
- Multiplication (resp. addition, division) of two elements of L_t can be performed with $O(\mathcal{M}(t))$ (resp. $O(t)$, $O(\mathcal{M}(t) \log t)$) L -operations; see [46, Corollary 11.8] for instance.

We split the proof of Theorems 3 and 4: (1) we study residue field representations, (2) we bound the number of field operations induced by changes of variables, (3) we estimate factorization costs, (4) we conclude.

5.1 Representation of residue fields and arithmetic in finite fields

RNPuiseux constructs residue fields k_i step by step by adding characteristic polynomial roots to the ground field. It would therefore be convenient to represent them as multiple extensions. More precisely:

$$k_i \simeq L[T_1, \dots, T_{m_i}]/(M_1(T_1), M_2(T_1, T_2), \dots, M_{m_i}(T_1, \dots, T_{m_i})),$$

for some integer m_i , where M_j is the minimal polynomial (with $\deg_{T_j}(M_j) \geq 2$) of a root over the previous coefficient field:

$$k_{i,j-1} \simeq L[T_1, \dots, T_{j-1}]/(M_1(T_1), \dots, M_{j-1}(T_1, \dots, T_{j-1})),$$

with $k_{i,0} = L$ and $k_{i,m_i} = k_i$. For $1 \leq j \leq m_i$, we define:

$$d_j = [k_{i,j} : k_{i,j-1}] \quad D_j = d_1 \cdots d_j = [k_{i,j} : L].$$

With such a triangular representation, it is shown in [29] that the number of operations in L necessary to perform an operation in k_i is bounded by $C4^{m_i} f_i \log^3(f_i)$, for some universal constant C . Since $m_i \leq \log_2 f_i$, we obtain $Cf_i^3 \log^3(f_i)$; with such a bound, we could not reach our goal, namely Theorem 3. Moreover, taming the “exponential” factor 4^{m_i} seems to be difficult [29].

Therefore, we turn to *primitive representations* for the fields $k_{i,j}$: $k_{i,j} \simeq L[T]/(P_j(T))$, where P_j is the minimal polynomial over L of a primitive element α_j , i.e. $k_{i,j} \simeq L(\alpha_j)$ for $1 \leq j \leq m_i$. Elements of $k_{i,j}$ are encoded as polynomials of degree less than D_j with coefficients in L . Whenever the root $\xi = \xi_j$ of a characteristic polynomial is not in the coefficient field, the following computations are required:

- Compute a primitive element α_j of $k_{i,j} = L(\xi_j, \alpha_{j-1})$ and its minimal polynomial P_j over L .
- Express ξ_j and α_{j-1} as polynomials in α_j of degree less than D_j with coefficients in L .
- Rewrite the coefficients of $H(X, Y) \in L(\alpha_{j-1})[X, Y]$ in terms of α_j .

Moreover, when a recursive call to `RNPuiseux` returns, note that the representation of ξ_j is different from the coefficient representation of the returned triplet $[G, P, Q]$. In order to form a meaningful result $\left[G, \xi_j^b P^q, P^m \left(\xi_j^a + Q\right)\right]$ we still need to:

(D) Rewrite ξ_j^a and ξ_j^b in terms of α_{m_i} .

These “backward” transformations must be executed each time the function returns. Since changes of representation are not required if multiple extensions are used, for the sake of simplicity, we have not included them in our description of `RNPuiseux`.

Let ω be a real number such that two square matrices of size n can be multiplied using $O(n^\omega)$ operations in their coefficient field. We assume $2 \leq \omega \leq 3$. The best known exponent is $\omega \approx 2.376$; see [14]. We emphasize that our complexity results do not require asymptotically fast matrix multiplication algorithms, which have so far demonstrated limited practical value. However, the analysis below show that some steps could be asymptotically sped up if such methods were used.

5.1.1 Step (A)

We choose a random linear combination of monomials in $\left\{\xi_j^l \alpha_{j-1}^m \mid 0 \leq l < d_j, 0 \leq m < D_{j-1}\right\}$ with coefficients in L and compute its minimal polynomial over L . If this polynomial has degree D_j , we are done; otherwise we repeat the process. It is well-known that the ratio of primitive elements of $k_{i,j}$ is bounded from below by a constant; see [40, Fact 5.8] for instance. Hence, the expected number of trials is in $O(1)$. To compute a minimal polynomial, we suggest two methods, both due to Shoup:

- (A₁) In [40], it is shown that such a minimal polynomial can be computed with $O\left(D_j^{(\omega+1)/2}\right)$ L -operations; see Theorem 3.4 and the subsequent remark therein. Writing $D_j^{(\omega+1)/2} \leq f_i d_Y^{(\omega-1)/2}$, multiplying by $(r_i - s_i)$ and summing over i , we obtain a total cost in $O\left(\delta_F d_Y^{(\omega-1)/2}\right)$. The method is based on the (theoretical) application of Tellegen’s transposition principle. If fast matrix multiplication is used, the method is asymptotically superior to the next one.
- (A₂) More recently, Shoup has given an explicit version of Tellegen’s principle for “power projections” that allows to compute minimal polynomials with $O\left(\mathcal{M}(D_j) D_j^{1/2} + D_j^2\right)$ L -operations; see [41]. The algorithm has been implemented by Shoup, who reported an experimental behaviour in accordance with the theory. Writing $\mathcal{M}(D_j) D_j^{1/2} + D_j^2 \leq 2f_i \widetilde{\mathcal{M}}(D_j) \mathcal{D}_j \leq 2f_i \mathcal{M}(d_Y)$, the total cost is $O(\delta_F \mathcal{M}(d_Y))$.

5.1.2 Step (B)

Again, we suggest two methods based on the work of Shoup:

- (B₁) A method in $O\left(D_j^{(\omega+1)/2}\right)$, leading to a total cost of $O\left(\delta_F d_Y^{(\omega-1)/2}\right)$, see [40, Theorem 3.5 and the subsequent remark therein]. This algorithm is based on

the following ingredients: The resolution of a Toeplitz system in $O(D_j \mathcal{L}(D_j) \log D_j) L$ -operations, the resolution of two power projection problems, and the evaluation of a univariate polynomial at an element of k_{ij} . The same comments apply as for $(\mathcal{A}_1)$.

- ($\mathcal{B}_2$) By [41], power projections and evaluation in $(\mathcal{B}_1)$ can be achieved by an explicit algorithm requiring $O\left(\mathcal{M}(D_j)D_j^{1/2} + D_j^2\right)$. This gives again $O(\delta_F \mathcal{M}(d_Y))$.

5.1.3 Step (C)

The number of coefficients of H that require a change of representation is bounded by $d_Y \delta_F / D_{j-1}$: Indeed, at the first function call, there is no need for a representation change, but for following calls, the truncation order is δ_F divided by the degree over L of the coefficient field. Each coefficient $C(\alpha_{j-1})$ can be viewed as a polynomial of degree less than D_{j-1} that must be evaluated at an element of k_{ij} . Three evaluation methods are considered. The first two ones are based on [5]; the last one is a naive approach. It is interesting to note that they all yield the same asymptotic bound. For practical purposes, the last two ones are preferable since they are the simplest ones.

- ($\mathcal{C}_1$) From [40, Fact 3.1], each evaluation can be done at a cost of

$$O\left(D_{j-1}^{\omega/2} \left[D_j / D_{j-1}^{1/2} + 1\right] + D_{j-1}^{1/2} \mathcal{M}(D_j)\right)$$

L -operations. Multiplying by $d_Y \delta_F / D_{j-1}$, simple calculations and inequalities lead to $O(\delta_F^2 \mathcal{M}(d_Y))$.

- ($\mathcal{C}_2$) Using the evaluation algorithm of [41], we get $O\left(\mathcal{M}(D_j)D_{j-1}^{1/2} + D_j D_{j-1}\right)$, hence $O(\delta_F^2 \mathcal{M}(d_Y))$ again.

- ($\mathcal{C}_3$) Finally, Horner scheme gives an evaluation cost of $D_{j-1} \mathcal{M}(D_j)$. Again, this gives $O(\delta_F^2 \mathcal{M}(d_Y))$.

We could have multiplied by $m_i \leq \log_2 f_i$ instead of $(r_i - s_i) \geq m_i$ and obtained $O(d_Y \delta_F \mathcal{M}(d_Y))$. However, we prefer to insist on the dependence on δ_F ; the result $O(\delta_F^2 \mathcal{M}(d_Y))$ cannot be deduced from the latter estimate since d_Y may be greater than δ_F , as demonstrated by the example $F(X, Y) = Y^n - X$ for $n > 2$.

5.1.4 Step (D)

For $2 \leq j \leq m_i$, let g_j be a univariate polynomial of degree less than D_j such that $\alpha_{j-1} = g_j(\alpha_j)$. Such polynomials have been computed at steps $(\mathcal{B})$. Using a backward induction argument, we show that a polynomial u_j such that $\alpha_j = u_j(\alpha_{m_i})$ may be computed: This is obviously the case for $j = m_i$. Writing $\alpha_{j-1} = g_j(u_j(\alpha_{m_i}))$, we see that u_{j-1} can be determined at the cost of a polynomial evaluation. Then, ξ_j^a and ξ_j^b can easily be rewritten in terms of α_{m_i} by polynomial evaluations since their expression in terms of α_j is known by $(\mathcal{B})$. Therefore, each step $(\mathcal{D})$ requires three evaluations of polynomials of degree less than f_i at elements of k_i , at a cost $O\left(f_i^{(\omega+1)/2}\right)$ (resp. $O\left(\mathcal{M}(f_i)f_i^{1/2} + f_i^2\right)$, $O(f_i \mathcal{M}(f_i))$) if approach $(\mathcal{C}_1)$ (resp.

(C_2) , (C_3)) is used. Multiplying by $(r_i - s_i)$ and summing over i , we get respectively $O\left(\delta_F d_Y^{(\omega-1)/2}\right)$, $O(\delta_F \mathcal{M}(d_Y))$ and $O(\delta_F \mathcal{M}(d_Y))$. To conclude step (D), remark that polynomials u_{j-1} could easily be returned by adding a fourth element to the triplet $\left[G, \xi_j^b P^q, P^m(\xi_j^a + Q)\right]$.

We have proved:

Proposition 6 *For an integer $N \geq \delta_F$, all steps of type (A), (B), (C) and (D) required by $\text{RNPuiseux}(L, F, N)$ can be performed with an expected number of $O(\delta_F^2 \mathcal{M}(d_Y))$ L -operations.*

Remark 4 The first author and Éric Schost have recently proposed a quasi-linear method to compute with triangular sets [36], based on remarkable results of Kedlaya and Umans [24, 25], that could improve estimates of this section and of Sects. 5.2 and 5.3. In this paper, we have decided not to take these results into account for three reasons: (1) it is not clear that they lead to an overall improvement of Theorem 3, (2) they do not seem to be of practical interest at this point, (3) they are of binary nature and do not yield estimates for the number of L -operations.

5.2 Changes of variables

Lemma 2 *Let N be a positive integer. Let $H \in L_t[X, Y]$ and $\xi \in L_t$ be as in RNPuiseux , with $\deg_Y(H) = d_Y$. Define $U(X, Y) = H(\xi^b X^q, X^m(\xi^a + Y)) / X^l$, where (m, q, l) corresponds to an edge Δ of a Newton polygon and (a, b) is given by algorithm Bézout. Then, the number of L_t -operations required to compute $\hat{H} = \lceil U \rceil^N$ is in:*

- $O(N \mathcal{M}(d_Y))$ if $\Delta \in \mathcal{GN}(H)$,
- $O(N \mathcal{M}(d_Y) + v(H))$ if $\Delta \in \mathcal{EN}(H)$, where $v(H) = v_X(\text{lc}_Y(H))$.

Proof First of all, note that points (i, j) of $\text{Supp}(H)$ along the line $mi + qj = w$ are transformed into points along the horizontal line $j = w - l$ by the change of variables; see Fig. 2.

Consider first the case $\Delta \in \mathcal{GN}(H)$. Since the slope is negative, we just need to perform a change of variables in $\tilde{H}(X, Y) = \sum_{w=l}^{N+l} H_w(X, Y)$ with $H_w(X, Y) =$

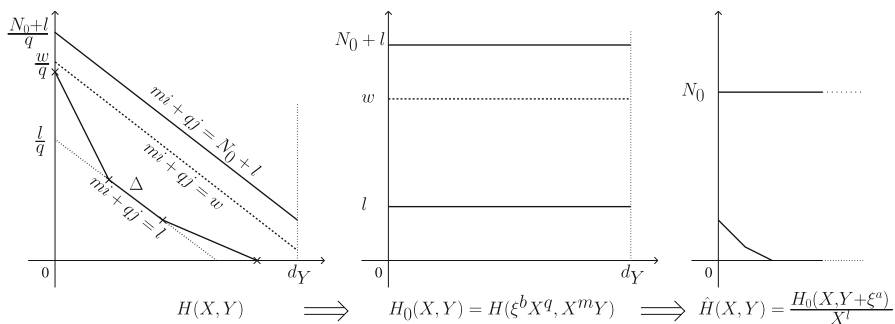


Fig. 2 Change of variables for negative slopes

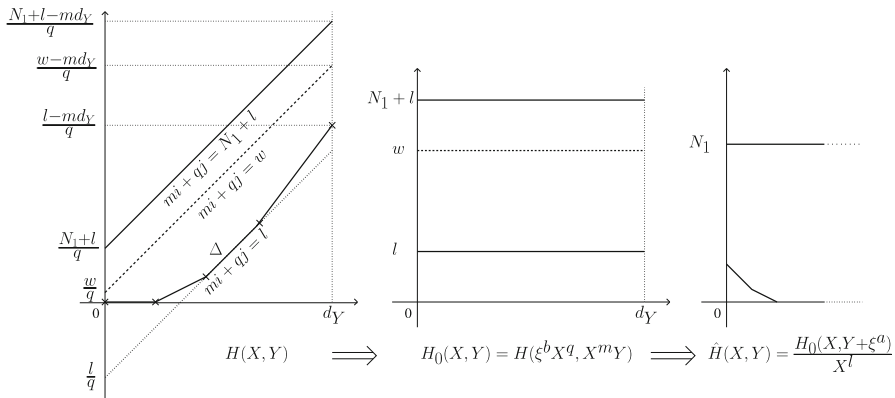


Fig. 3 Change of variables for positive slopes

$\sum_{mi+qj=w} \alpha_{ij} X^j Y^i$. We have:

$$\begin{aligned} H_w(\xi^b X^q, \xi^a X^m (1+Y)) &= \sum_{mi+qj=w} \alpha_{ij} (\xi^b X^q)^j (X^m (\xi^a + Y))^i \\ &= X^w \sum_{mi+qj=w} \alpha_{ij} \xi^{bj} (\xi^a + Y)^i \\ &= X^w V_w(Y + \xi^a) \end{aligned}$$

where $V_w(Z) = \sum_{mi+qj=w} \alpha_{ij} \xi^{bj} Z^i$ is a polynomial of degree at most d_Y . If a and b are chosen as in Bézout, $0 \leq b < q \leq d_Y$ and $|a| \leq m$. Since $m/q \leq 1$, ξ^a and ξ^b can be computed with $O(\log_2 d_Y) L_t$ -operations using “square and multiply” technique. We then form relevant powers of ξ^b . The exponent j is bounded by $(N+l)/q$, as illustrated by Fig. 2. Since slopes of generic Newton polygons are at least -1 , we have $l/q \leq d_Y/q \leq d_Y$; computing all powers up to this bound is achieved in $O(N+d_Y) L_t$ -operations. Constructing the polynomials V_w requires at most $N d_Y$ multiplications in L_t . Finally, since $p > d_Y$, a shift in V_w can be reduced to the multiplication of two polynomials of degree at most d_Y , with cost $O(\mathcal{M}(d_Y))$; see [1, Problem 2.6]. Since we must perform N such shifts, the total cost is in $O(\mathcal{M}(d_Y)N)$.

For $\Delta \in \mathcal{EN}(H)$, we proceed similarly. It is easily seen that $|a| \leq |m| \leq v(H)$, so that ξ^b may be computed with $O(\log_2 v(H)) L_t$ -operations. The largest exponent j for powers of ξ^b is now at most $(N+l-md_Y)/q$; see Fig. 3. From $(l-md_Y)/q \leq v(H)$, we get a number of operations in $O(N+v(H))$. To complete the proof, remark that the number of H_w is the same as before.

Proposition 7 *Let $N \geq \delta_F$ be an integer. Changes of variables induced by $\text{RNPuiseux}(L, F, N)$ can be performed with a number of L -operations in:*

$$O([N \eta_F \mathcal{M}(d_Y) + v d_Y] \widetilde{\mathcal{M}}(d_Y) \log d_Y)$$

Proof To compute the RPE R_i , RNPuiseux performs at most $(r_i - s_i + 1)$ changes of variables over a field included in $L_{f_i} = k_i$, the first one corresponding to the exceptional Newton polygon $\mathcal{EN}(F)$. Taking into account the extension L_{f_i}/L and Lemma 2, the number of L -operations required by the changes of variables for the i th RPE is in:

$$O([N \mathcal{M}(d_Y)(r_i - s_i + 1) + v] \mathcal{M}(f_i) \log f_i).$$

Writing $\mathcal{M}(f_i) \log f_i \leq f_i \widetilde{\mathcal{M}}(d_Y) \log d_Y$ and summing over i yields the proposition, considering Theorem 1.

5.3 Factorization cost

If N and t are positive integers, define:

$$\mathcal{F}(N, t) = \mathcal{M}(N^2) \log N + t \mathcal{M}(N) \log N \log p.$$

The factorization over $\mathbb{F}_{p'}$ of a univariate polynomial of degree N in $\mathbb{F}_{p'}[T]$ can be determined with an expected number of $O(\mathcal{F}(N, t))$ $\mathbb{F}_{p'}$ -operations; see [46, Corollary 14.30].

Remark 5 There exist algorithms with better complexity bounds; see [23, 47] for surveys and [25] for recent results. However, we have preferred to stick to well established algorithms, since factorization cost, for a fixed value of p , does not dominate our complexity bound.

Proposition 8 *The expected number of L -operations required to factorize all characteristic polynomials appearing in $\text{RNPuiseux}(L, F, \delta_F)$ is in:*

$$O((\delta_F + 1) \mathcal{F}(d_Y, t_0) \mathcal{L}(d_Y) \log d_Y) \subset O^* \left((\delta_F + 1) \left[\mathcal{M}(d_Y^2) + t_0 \log p \mathcal{M}(d_Y) \right] \right).$$

Proof Let L_t be the extension of L over which a factorization of a characteristic polynomial ϕ_Δ must be determined. First of all, we prove that the degree d_Δ of ϕ_Δ is at most d_Y/t . This is obviously true at the first stage of the algorithm, where $t = 1$. Assume that the property is true at a given stage of the algorithm and denote $H \in L_t[X, Y]$ the input polynomial, with an edge Δ . Let ξ be a root of ϕ_Δ and k be its multiplicity in ϕ_Δ . At the next function call, let $\phi_{\Delta'}$ be a characteristic polynomial of the polygon yielded by this choice of Δ and ξ ; denote its degree by $d_{\Delta'}$. Since k is the number of Puiseux series of H having $\xi X^{m_\Delta/q_\Delta}$ as initial term, necessarily, $d_{\Delta'} \leq k$. But $k[L_t(\xi) : L_t] \leq d_\Delta$ and $d_\Delta \leq d_Y/t$ by the induction hypothesis. Therefore $d_{\Delta'} \leq d_Y/[L_t(\xi) : L]$; this proves the property. Hence, factorization of a characteristic polynomial can be achieved with an average number of $O\left(\mathcal{F}\left(\frac{d_Y}{t}, t_0 t\right)\right)$ L_t -operations. We must now multiply by the cost of each L_t -operations, namely $O(\mathcal{M}(t) \log t) L$ -operations. By property (10), we have $\mathcal{M}\left(\left(\frac{d_Y}{t}\right)^2\right) \log \frac{d_Y}{t} \mathcal{M}(t) \log t \in O(\mathcal{M}(d_Y^2) \mathcal{L}(d_Y) \log^2 d_Y)$

and $\mathcal{M}\left(\frac{d_Y}{t}\right) \log \frac{d_Y}{t} \mathcal{M}(t) \log t \in O\left(\mathcal{M}(d_Y) \mathcal{L}(d_Y) \log^2 d_Y\right)$, no matter which arithmetic is used. Hence, the number of L -operations belongs to:

$$O\left(\mathcal{F}(d_Y, t_0 t) \mathcal{L}(d_Y) \log d_Y\right). \quad (12)$$

We treat separately characteristic polynomials of $\mathcal{EN}(F)$. Remark that the sum of their degree is less than d_Y . Property (11) extends to $\mathcal{F}$; therefore, the total cost of factoring characteristic polynomials of $\mathcal{EN}(F)$ is in $O(\mathcal{F}(d_Y, t_0))$. For other characteristic polynomials, we multiply estimate (12) by $r_i - s_i$, bound t by f_i and sum over i to obtain the result. Note that, since δ_F may be null, factorization cost for $\mathcal{EN}(F)$ must be explicitly added, yielding the factor $\delta_F + 1$ of the bound.

5.4 Proof of Theorems 3 and 4

In this section, we use asymptotically fast algorithms for multiplication of univariate polynomials. It is interesting to bound first the number of L -operations in terms of quantities related to the output size, namely η_F and δ_F .

Lemma 3 *Let $N \geq \delta_F$ be an integer. The expected number of L -operations required by $\text{RNPuiseux}(L, F, N)$ belongs to:*

$$O^\sim \left(d_Y \left[\eta_F N + v + \delta_F^2 + (\delta_F + 1)(d_Y + t_0 \log p) \right] \right).$$

Proof Follows easily from Propositions 6, 7 and 8;

Lemma 4 *Define $\alpha_F = v_X(R_F)$. There is an algorithm to compute singular parts of all RPEs of F above 0 with an expected number of L -operations in:*

$$O^\sim \left((\alpha_F + 1) d_Y \left[\alpha_F + d_Y + t_0 \log p \right] + v d_Y \right)$$

Proof By Proposition 5, $\text{RNPuiseux}(L, F, \alpha_F)$ returns the expected output. Then, apply Lemma 3 with $\delta_F \leq \alpha_F$ and $\eta_F \leq \alpha_F + d_Y$. Note that α_F may be null, so that the factor $\alpha_F + 1$ cannot be simplified.

Proof of Theorem 3: Consequence of Lemma 4 since $\alpha_F \leq d_X(2d_Y - 1)$.

We now turn to Theorem 4.

Lemma 5 *Define $\mu_F = \deg_X R_F$. There exists an algorithm to compute singular parts of all RPEs of F above all affine critical points with an expected number of L -operations in:*

$$O^\sim \left(d_Y \mu_F \left[d_Y + d_X + \mu_F t_0 \log p \right] + d_X d_Y^2 \right).$$

Proof To deal with *all* conjugacy classes over L of affine critical points, we proceed as follow: If $R_F = c \prod_i \Phi_i^{h_i}$ is a factorization of R_F into a product of monic irreducible polynomials $\Phi_i \in L[X]$, RPEs above roots of Φ_i are conjugated over L . Therefore, it is sufficient to compute a system of RPEs above one root c_i of Φ_i for each i .

The calculation of R_F can be done with $O^{\sim}(d_X d_Y^2) L$ -operations; see [46, Corollary 11.18, Algorithm 6.59] if L contains at least $(6d_Y + 3)d_X$ elements, or see [37] otherwise. Thus, this step is included in our complexity bound.

Then, R_F can be factorized over L in $O(\mathcal{F}(\mu_F, t_0)) \subset O^{\sim}(\mu_F^2 + \mu_F t_0 \log p)$ L -operations; see [46, Corollary 14.30]. This step is also included in our complexity bound.

Coefficients of $F_i = F(X + c_i, Y)$ can be computed at the cost of d_Y shifts in the coefficients of F in Y ; the complexity of this step is in $O^{\sim}(d_Y d_X)$ field operations in $L_{t_i} = L[T]/(\Phi_i(T))$ using a “divide and conquer” approach for shifts.⁴ Set $t_i = \deg_X(\Phi_i)$. In terms of L -operations, the cost is therefore in $O^{\sim}(d_Y d_X t_i)$. Summing over i and bounding $\sum_i t_i$ by μ_F gives again a sufficient estimate.

Define $v_i = v_X(\text{lc}_Y(F_i))$. By Lemma 4 with $\alpha_{F_i} = h_i > 0$, RPEs of F above $X = c_i$ may be computed using an expected number of L_{t_i} -operations in:

$$O^{\sim}(h_i d_Y [h_i + d_Y + t_i t_0 \log p + v_i]) \subset O^{\sim}(h_i d_Y [\mu_F + d_Y + \mu_F t_0 \log p + d_X]).$$

Multiplying by t_i to get the number of operations in L and summing over i allows to conclude since $\sum_i h_i t_i = \mu_F$.

Proof of Theorem 4: For affine critical points, apply Lemma 5 with $\mu_F \leq d_X(2d_Y - 1)$. For RPEs above infinity, define $G(X, Y) = F(1/X, Y)X^{d_X} \in L[X, Y]$; G satisfies hypotheses of RNPuiseux input. Obviously, $\deg_X(G) = d_X$ and $\deg_Y(G) = d_Y$. To complete the proof, apply Theorem 3 to G .

6 Conclusion

We have presented an $O^{\sim}(d^5 \log q)$ algorithm to compute singular parts of rational Puiseux expansions of a total degree d bivariate polynomial over a field with q elements. The genus of the curve defined by this polynomial may be determined at the same cost. We conclude this paper with a number of remarks:

Improving the arithmetic in a degree t finite field represented as a multiple extension to achieve an $O(M(t))$ complexity would render Sect. 5.1 obsolete.

Faster algorithms could be used for subproblems such as factorization and polynomial evaluation. It is not clear that this would improve our main result: Indeed, the dominating term would be $O^{\sim}(d_Y \delta_F \eta_F)$ anyway (Proposition 7 with $N = \delta_F$), since we have been unable to exhibit an a priori bound better than $O(d_X d_Y)$ for δ_F and η_F . For sparse cases, the theory of mixed volumes of Newton polytopes is likely to provide finer results.

⁴ Note that the method based on a reduction to a polynomial multiplication requires $p > d_X$ and cannot be applied.

Algorithm `RNPuiseux` returns truncated series in “Horner-like form”; see examples in Sect. 3. If expanded forms are required, it is easy to check that they can be obtained with $O(\eta_F^2)$ operations in L .

Finally, note that our bounds, combined with estimates for the size of a good prime p [35], yield bit-complexity results when the ground field K is an algebraic number field. Define $K = \mathbb{Q}(\gamma)$ and let $M_\gamma \in \mathbb{Z}[T]$ be the minimal polynomial of γ over $\mathbb{Q}$. Suppose that $F \in K[X, Y]$ defines an irreducible curve $F(X, Y) = 0$. If P is a multivariate polynomial in $K[\underline{X}]$, let (H, c) be the unique pair in $H \in \mathbb{Z}[T, \underline{X}]$, $c \in \mathbb{N}$, with $\deg_T(H) < w$ and $P(\underline{X}) = H(\gamma, \underline{X})/c$, where c is minimal. We define $\text{ht}(P) = \max\{\log c, \log \|H\|_\infty\}$.

From [35, Proposition 58] or [32, Théorème 17], we can easily deduce:

Theorem 5 *Let ϵ be a real number with $0 < \epsilon \leq 1$. There exists a Monte Carlo algorithm that computes the genus of the curve $F(X, Y) = 0$ with probability of error less than ϵ and an expected number of word operations in:*

$$O^-(d_Y^3 d_X^2 w^2 \log^2 \epsilon^{-1} [\text{ht}(M_\gamma) + \text{ht}(F)]) .$$

From [35, Proposition 59] or [32, Théorème 18] we get:

Theorem 6 *Assuming that the degree of the squarefree part of the resultant R_F is known, there exists a Las Vegas algorithm that computes the genus of the curve $F(X, Y) = 0$ with an expected number of word operations in:*

$$O^-(d_Y^3 d_X^2 w^2 [\text{ht}(M_\gamma) + \text{ht}(F)]) .$$

Unfortunately, an algorithm for the computation of the squarefree part of R_F with a complexity that fits into this estimate is unknown to the authors, due to coefficient growth. A deterministic choice of p based on [35, Proposition 56] yields an algorithm with average running time in $O^-(d^{15} w^4 [\text{ht}(M_\gamma) + \text{ht}(F)]^2)$. This is not encouraging, but should be compared to Walsh’s estimate $O(d^{36+\epsilon} \text{ht}(F)^{2+\epsilon})$ for the computation of CPS of $F \in \mathbb{Q}[X, Y]$ above 0 [50].

Acknowledgments We thank an anonymous referee for suggestions to improve the readability of the paper.

References

1. Bini, D., Pan, V.Y.: Polynomial and Matrix Computations, Progress in Theoretical Computer Science. Vol. 1, Birkhäuser, Saarbrücken (1994)
2. Bliss, G.A.: Algebraic functions. AMS (1933)
3. Bosma, W., Cannon, J., Playoust, C.: The Magma algebra system I : the user language. J. Symb. Comput. **24**(3–4), 235–265 (1997)
4. Bostan, A., Chyzak, F., Lecerf, G., Salvy, B., Schost, E.: Differential equations for algebraic functions. In: Brown, C.W. (ed.) ISSAC’07: Proceedings of the 2007 International Symposium on Symbolic and Algebraic Computation, pp. 25–32. ACM Press (2007). doi:[10.1145/1277548.1277553](https://doi.org/10.1145/1277548.1277553)
5. Brent, R.P., Kung, H.T.: Fast algorithms for manipulating formal power series. J. ACM. **25**(4), 581–595 (1978). doi:[10.1145/322092.322099](https://doi.org/10.1145/322092.322099)

6. Brieskorn, E., Knörrer, H.: *Plane Algebraic Curves*. Birkhäuser (1986)
7. Cantor, D., Kaltofen, E.: On fast multiplication of polynomials over arbitrary algebras. *Acta Inform.* **28**(7), 693–701 (1990)
8. Chevalley, C.: *Introduction to the Theory of Algebraic Functions of One Variable*, Mathematical Surveys, vol. 6. AMS (1951)
9. Chistov, A.L.: Polynomial complexity of the Newton-Puiseux algorithm. In: *Mathematical Foundations of Computer Science 1986*, pp. 247–255. Springer, London, UK (1986)
10. Chudnovsky, D.V., Chudnovsky, G.V.: On expansion of algebraic functions in power and Puiseux series I. *J. Complex.* **2**(4), 271–294 (1986)
11. Chudnovsky, D.V., Chudnovsky, G.V.: On expansion of algebraic functions in power and Puiseux series II. *J. Complex.* **3**(1), 1–25 (1987)
12. Cohn, P.M.: Puiseux's theorem revisited. *J. Pure Appl. Algebra* **24**, 1–4 (1984)
13. Comtet, L.: Calcul pratique des coefficients de Taylor d'une fonction algébrique. *L'Enseignement Mathématique* **2**(10), 267–270 (1964)
14. Coppersmith, D., Winograd, S.: Matrix multiplication via arithmetic progressions. In: *STOC '87: Proceedings of the Nineteenth Annual ACM Symposium on Theory of Computing*, pp. 1–6. ACM, New York, NY, USA (1987). doi:[10.1145/28395.28396](https://doi.org/10.1145/28395.28396)
15. Cormier, O., Singer, M.F., Trager, B.M., Ulmer, F.: Linear differential operators for polynomial equations. *J. Symbol. Comput.* **34**(5), 355–398 (2002)
16. Dahan, X., Schost, E., Maza, M.M., Wu, W., Xie, Y.: On the complexity of the D5 principle. *SIGSAM Bull.* **39**(3), 97–98 (2005). doi:[10.1145/1113439.1113457](https://doi.org/10.1145/1113439.1113457)
17. Della Dora, J., Dicrescenzo, C., Duval, D.: About a new method for computing in algebraic number fields. In: *EUROCAL 85*. Springer-Verlag LNCS 204 (1985)
18. Diaz-Toca, G., Gonzalez-Vega, L.: Determining Puiseux expansions by Hensel's lemma and dynamic evaluation. In: Ganzha, V., Mayr, E., Vorozhtsov, E. (eds.) *Computer Algebra in Scientific Computing, CASC 2002*, Technische Universität München, Germany (2002)
19. Duval, D.: *Diverses questions relatives au calcul formel avec des nombres algébriques*. Université de Grenoble, Thèse d'État (1987)
20. Duval, D.: Rational Puiseux expansions. *Compos. Math* **70**(2), 119–154 (1989)
21. Eichler, M.: *Introduction to the Theory of Algebraic Numbers and Functions*. Academic Press, London (1966)
22. Henry, J.P., Merle, M.: Complexity of computation of embedded resolution of algebraic curves. In: *Proceedings Eurocal 87*, no. 378 in *Lecture Notes in Computer Science*, pp. 381–390. Springer (1987)
23. Kaltofen, E.: Polynomial factorization: a success story. In: *ISSAC '03: Proceedings of the 2003 International Symposium on Symbolic and Algebraic Computation*, pp. 3–4. ACM, New York, NY, USA (2003). doi:[10.1145/860854.860857](https://doi.org/10.1145/860854.860857)
24. Kedlaya, K., Umans, C.: Fast modular composition in any characteristic. In: *FOCS*, pp. 146–155. IEEE Computer Society (2008)
25. Kedlaya, K.S., Umans, C.: Fast polynomial factorization and modular composition. In: *STOC*, pp. 481–490 (2008)
26. Kung, H.T., Traub, J.F.: All algebraic functions can be computed fast. *J. ACM* **25**(2), 245–260 (1978)
27. Lang, S.: *Algebra*. Addison-Wesley, Reading (1965)
28. Lecerf, G.: Fast separable factorization and applications. *Appl. Algebra Eng. Commun. Comput.* **19**(2), (2008)
29. Li, X., Maza, M.M., Schost, E.: Fast arithmetic for triangular sets: from theory to practice. In: *ISSAC '07: Proceedings of the 2007 International Symposium on Symbolic and Algebraic Computation*, pp. 269–276. ACM, New York, NY, USA (2007) doi:[10.1145/1277548.1277585](https://doi.org/10.1145/1277548.1277585)
30. Monagan, M.B., Geddes, K.O., Heal, K.M., Labahn, G., Vorkoetter, S.M., McCarron, J., DeMarco, P.: *Maple 10 Programming Guide*. Maplesoft, Waterloo (2005)
31. Poteaux, A.: Computing monodromy groups defined by plane algebraic curves. In: *Proceedings of the 2007 International Workshop on Symbolic-Numeric Computation*, pp. 36–45. ACM, New-York (2007)
32. Poteaux, A.: *Calcul de développements de Puiseux et application au calcul de groupe de monodromie d'une courbe algébrique Plane*. Ph.D. thesis, Université de Limoges (2008)
33. Poteaux, A., Rybowicz, M.: Good reduction of Puiseux series and complexity of the Newton-Puiseux algorithm. In: *ISSAC '08: Proceedings of the 2008 International Symposium on Symbolic and Algebraic Computation*, pp. 239–246. ACM, New-York (2008)

34. Poteaux, A., Rybowicz, M.: Towards a Symbolic-Numeric Method to Compute Puiseux Series: The Modular Part. <http://arxiv.org/abs/0803.3027> (2008)
35. Poteaux, A., Rybowicz, M.: Good reduction of Puiseux series and applications. *J. Symbol. Comput.* (To appear, 2011)
36. Poteaux, A., Schost, E.: Modular composition modulo triangular sets and applications. *Comput. Complex.* (Submitted, 2010)
37. Reischert, D.: Asymptotically fast computations of subresultants. In: ISSAC '97: Proceedings of the 1997 International Symposium on Symbolic and Algebraic Computation, pp. 233–240. ACM, ACM Press (1997)
38. Sasaki, T., Inaba, D.: Hensel construction of $f(x, u_1, \dots, u_l)$ at a singular point and its application. *Sigsam Bull.* **1**, 9–17 (2000)
39. Schönage, A., Strassen, V.: Schnelle multiplikation großer zahlen. *Comput.* **7**, pp. 281–292 (1971)
40. Shoup, V.: Fast construction of irreducible polynomials over finite fields. *J. Symbol. Comput.* **17**, 371–391 (1993)
41. Shoup, V.: Efficient computation of minimal polynomials in algebraic extensions of finite fields. In: IS-SAC '99: Proceedings of the 1999 International Symposium on Symbolic and Algebraic Computation, pp. 53–58. ACM, New York, NY, USA (1999) doi:[10.1145/309831.309859](https://doi.org/10.1145/309831.309859)
42. Teitelbaum, J.: The computational complexity of the resolution of plane curve singularities. *Math. Comp.* **54**(190), 797–837 (1990)
43. van der Hoeven, J.: Fast evaluation of holonomic functions. *Theor. Comput. Sci.* **210**(1), 199–215 (1999)
44. van der Hoeven, J.: Effective analytic functions. *J. Symbol. Comput.* **39**(3–4), 433–449 (2005)
45. van Hoeij, M.: An algorithm for computing an integral basis in an algebraic function field. *J. Symbol. Comput.* **18**, 353–363 (1994)
46. von zur Gathen, J., Gerhard, J.: *Modern Computer Algebra*. Cambridge University Press, Cambridge (1999)
47. Von Zur Gathen, J., Panario, D.: Factoring polynomials over finite fields: a survey. *J. Symbo. Comput.* **31**(1–2), 3–17 (2001). doi:[10.1006/jsco.1999.1002](https://doi.org/10.1006/jsco.1999.1002)
48. Walker, R.J.: *Algebraic Curves*. Springer, Berlin (1978)
49. Walsh, P.G.: On the complexity of rational Puiseux expansions. *Pac. J. Math.* **188**, 369–387 (1999)
50. Walsh, P.G.: A polynomial-time complexity bound for the computation of the singular part of an algebraic function. *Math. Comput.* **69**, 1167–1182 (2000)
51. Yun, D.D.Y.: On square-free decomposition algorithms. In: *Proceedings SYMSAC '76*. ACM (1976)
52. Zariski, O.: *Le problème des modules pour les branches planes*. Hermann, Paris (1981)